Fairness in Medical Federated Learning: A Layer-Wise Re-Weighting Approach (Fed-LWR)

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***Abstract*—Federated Learning (FL) enables hospitals to collaboratively train models without sharing raw patient data, providing strong privacy guarantees. However, widely used aggregation schemes such as FedAvg optimize only global accuracy and overlook fairness across clients, resulting in unstable performance when hospitals exhibit severe domain shift. This paper introduces Fed-LWR, a novel fairness-aware FL framework that performs layer-wise re-weighting by measuring CKA-based representational similarity between local and global feature spaces. By adaptively assigning larger influence to clients whose representations deviate more from the global anchor, Fed-LWR achieves a significantly improved fairness–accuracy trade-off and ensures that heterogeneous institutions contribute meaningfully to the global update.**

We further propose Hier-Fed-LWR, a multi-level extension that integrates regional aggregation to better model large-scale medical networks involving multiple hospitals and acquisition protocols. This hierarchical design enhances stability under strong domain shift, reduces cross-client variance, and improves scalability for real deployments. Experiments on prostate MRI and retinal fundus datasets demonstrate that our methods consistently outperform existing fairness-oriented FL baselines, offering faster convergence, lower variance, and more equitable performance across domains.

***Index Terms*—Federated Learning, Fairness, Medical Imaging, Hierarchical FL, CKA Similarity, Layer-wise Aggregation**

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I. Introduction

Federated Learning (FL) enables collaboration across hospitals without sharing raw data, allowing institutions to leverage diverse datasets to train stronger global models while preserving patient privacy. In medical contexts this is particularly valuable: hospitals operate different scanner models, acquisition pipelines, and have distinct patient demographics, all of which lead to heterogeneity in data distributions.

However, standard FL schemes such as FedAvg optimize aggregate performance and often neglect fairness across participants. When domain shifts exist, some hospitals may benefit significantly while others see degraded performance, which reduces incentives to participate and undermines sustainable collaboration. Ensuring that all hospitals benefit — *performance fairness* — is therefore essential for real-world clinical FL deployments.

This paper proposes **Fed-LWR** (Federated Layer-Wise Re-weighting), a representation-aware aggregation strategy that measures layer-wise representational distances between local client models and a global anchor using Centered Kernel Alignment (CKA). These distances are converted to layer-specific weights that guide aggregation, improving fairness while preserving or improving average performance. We also present **Hier-Fed-LWR**, a hierarchical multi-level extension that inserts regional aggregators between hospitals and the global server to improve stability and scalability in large federations.

II. Related Work and Limitations of Existing Fairness Approaches

A number of methods have been proposed to improve fairness or personalization in FL. Personalization techniques like Ditto introduce client-specific objectives and penalties to adapt global models locally. Re-weighting approaches such as FedCE compute client contributions from gradient-based or validation-based signals. Other methods assign a single scalar weight per client using empirical loss or held-out validation metrics.

While effective in some settings, these approaches share two limitations in medical imaging:

1) **Coarse client-level weighting:** Assigning a single weight per client ignores the fact that domain shift often manifests differently across layers of a deep network. Early convolutional filters may remain aligned across hospitals, while deeper semantic layers diverge.
2) **Limited robustness to layer-wise distribution shifts:** Methods that do not account for layer-dependent representation mismatch risk suboptimal aggregation, poorer fairness, and oscillatory optimization behavior.

These shortcomings motivate a layer-wise approach that compares representations at each layer and adapts aggregation accordingly.

III. Preliminaries

A. Federated Learning Setup

Consider K hospitals participating in federated training through a trusted central server. Hospital $k \in \{1, \ldots, K\}$ holds a private dataset $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}$. Each site trains a deep neural network $f = h^{(M)} \circ \cdots \circ h^{(1)}$ with M layers. The standard federated objective is to minimize local empirical risk L_k at each client and aggregate client updates to form a global model:

$$w_t^G = \frac{1}{K} \sum_{k=1}^{K} w_t^k,$$

where w_t^G and w_t^k are the global and local model parameters at round t.

B. Performance Fairness

We call model $\hat{w}$ *fairer* than w if its performance (e.g., Dice score) is more uniform across the K hospitals. Average accuracy alone is insufficient — a model should avoid disproportionately advantaging some hospitals while disadvantaging others.

C. Dice Score

For segmentation tasks, we use the Dice coefficient:

$$\text{Dice}(X, Y) = \frac{2|X \cap Y|}{|X| + |Y|},$$

which ranges from 0 (no overlap) to 1 (perfect overlap). Dice is sensitive to false positives and negatives and is widely used in medical image segmentation.

IV. Hilbert-Schmidt Independence Criterion and CKA

A. HSIC

The Hilbert–Schmidt Independence Criterion (HSIC) measures dependence between two feature sets. Given centered Gram matrices K and L (computed from features $U \in \mathbb{R}^{n \times p}$ and $V \in \mathbb{R}^{n \times q}$), and centering matrix $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, the empirical HSIC is

$$\text{HSIC}(K, L) = \frac{1}{(n-1)^2}\text{tr}(KHLH).$$

Higher HSIC indicates stronger dependence / similarity between representations.

B. Centered Kernel Alignment (CKA)

CKA normalizes HSIC to provide a scale-invariant similarity score between representations:

$$\text{CKA}(U, V) = \frac{\text{HSIC}(K, L)}{\sqrt{\text{HSIC}(K, K) \cdot \text{HSIC}(L, L)}}.$$

CKA lies in $[0, 1]$: 1 denotes identical representations and 0 denotes dissimilar representations. We use CKA to compare layer-wise representations between a client's local model and the global anchor.

V. Fed-LWR: Layer-wise Re-weighting

A. Intuition

We observe that domain shift in medical imaging tends to be layer-dependent. Fed-LWR uses the global model as an *anchor* and computes layer-wise CKA between local and global representations. Large representational deviation at a layer indicates that the global model is far from a client's local optimum there; such clients should have more influence at that layer to improve fairness.

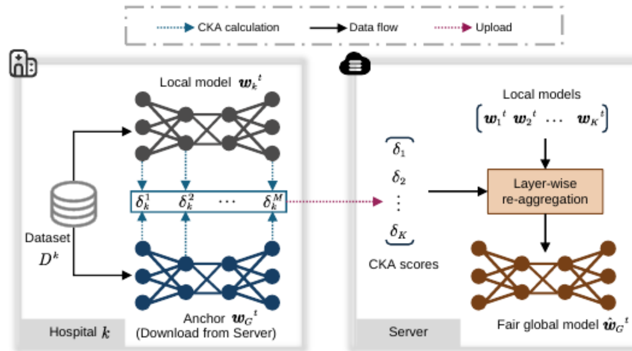


Fig. 1: Overview of Fed-lwr

B. Representation Difference Estimation

After local training in round t, the global anchor w_t^G is sent to each hospital. For hospital k and layer m, we compute:

$$\delta_k^m = \text{CKA}\big(U_k^m, V_k^m\big),$$

where U_k^m (resp. V_k^m) are the feature matrices produced by the local (resp. global) encoder at layer m on a shared set of inputs (or summary batches). The per-client layer-wise similarity vector is

$$\delta_k = [\delta_k^1, \ldots, \delta_k^M].$$

C. Layer-wise Weight Computation

We convert similarity to a weight that emphasizes larger representational gaps:

$$\rho_k^m = \frac{1 - \delta_k^m}{\sum_{i=1}^{K}(1 - \delta_i^m)}.$$

Hospitals with lower CKA (larger $1 - \delta$) receive higher weight at layer m.

D. Layer-wise Aggregation

Each layer of the global model is aggregated separately:

$$\hat{w}_{G,t}^m = \sum_{k=1}^{K} \rho_k^m \, w_{k,t}^m,$$

and the full global model is reconstructed from the aggregated per-layer parameters $\hat{w}_{G,t} = \{\hat{w}_{G,t}^1, \ldots, \hat{w}_{G,t}^M\}$.

E. Algorithm

VI. Experiments

A. Datasets

We evaluate on two medical segmentation benchmarks:

- **ProstateMRI**: six hospital partitions, images resized to 384×384.
- **RIF Fundus Imaging (RIF)**: four hospital domains with substantial domain heterogeneity; images resized to 384×384.

Train/validation/test splits follow the dataset definitions (ProstateMRI: 6:2:2; RIF: provided testing split and 4:1 training split).

Algorithm 1 Fed-LWR

Require: K hospitals, communication rounds T, learning rate η
Ensure: $\hat{w}_T^G$
Initialize w_0^G
for $t = 1$ to T **do**
  for each hospital $k = 1$ to K **in parallel do**
    $w_t^k \leftarrow \hat{w}_{t-1}^G$
    Local update: $w_t^k \leftarrow w_t^k - \eta \nabla L_k$
  end for
  Anchor aggregation: $w_t^G \leftarrow \frac{1}{K} \sum_{k=1}^K w_t^k$
  for each hospital $k = 1$ to K **in parallel do**
    Compute layer-wise similarities $\delta_k^m = \mathrm{CKA}(U_k^m, V_k^m)$ for $m = 1, \ldots, M$
  end for
  Compute layer-wise weights $\rho_k^m = \dfrac{1 - \delta_k^m}{\sum_{i=1}^K (1 - \delta_i^m)}$
  for each layer $m = 1$ to M **do**
    Aggregate: $\hat{w}_{G,t}^m \leftarrow \sum_{k=1}^K \rho_k^m \, w_{k,t}^m$
  end for
end for
return $\hat{w}_T^G$

B. Baselines

We compare Fed-LWR to:

- **Solo:** clients train independently (no communication).
- **FedAvg:** standard federated averaging.
- **q-FedAvg:** re-weighting via empirical loss.
- **CFFL:** server reallocation using validation contributions.
- **CGSV:** contribution estimation via cosine-gradient Shapley values.
- **Ditto:** personalized objectives with per-client optimization.
- **FedCE:** gradient- and validation-based client re-weighting (sum/multi variants).

C. Metrics

We use Dice coefficient for segmentation accuracy and report the average and standard deviation across clients to evaluate fairness and stability.

VII. Results

A. ProstateMRI

Table I shows test Dice scores per site. Fed-LWR attains the highest average Dice and the lowest inter-site standard deviation, demonstrating improved fairness and accuracy.

TABLE I: ProstateMRI Dice Scores

Method	Avg Dice (%)	Std (%)
FedAvg	88.78	4.54
FedCE (Multi)	91.57	1.65
Fed-LWR	**93.25**	**0.97**

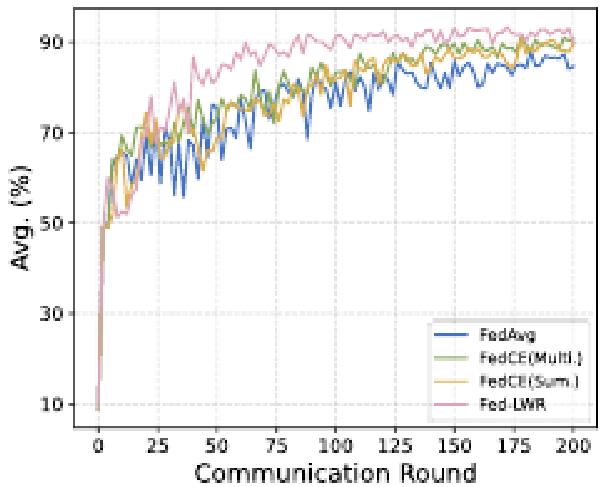


Fig. 2: Average Dice vs. communication rounds on the ProstateMRI testing set.

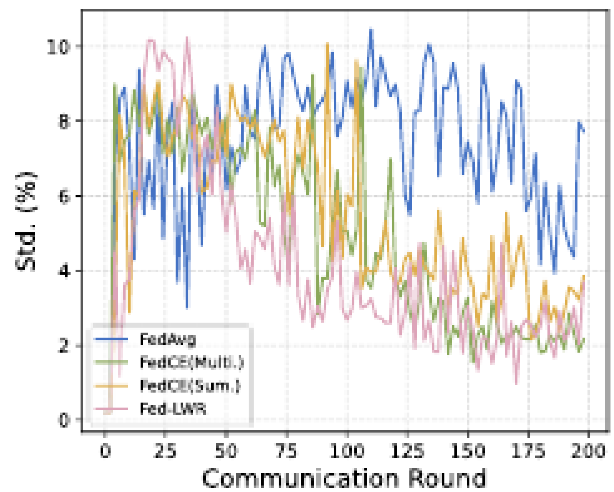


Fig. 3: Standard deviation across hospitals vs. communication rounds (ProstateMRI).

B. RIF Fundus Imaging

On RIF, Fed-LWR consistently improves fairness and average performance relative to baselines, demonstrating the benefit of layer-wise aggregation under strong domain shift.

C. Ablation Study

We evaluate two ablations:

- **Fed-LWR-v1:** cosine similarity in place of CKA.
- **Fed-LWR-v2:** aggregation using only the last encoder layer's similarity.

Results show that CKA and full layer-wise aggregation are critical: cosine performs worse and last-layer-only aggregation underperforms full Fed-LWR, confirming our design choices.

D. Convergence Analysis

Fed-LWR converges quickly and stabilizes around $\sim$90 communication rounds. We observe larger variance between rounds 20–50 due to adaptive fairness adjustments; this reflects active re-weighting rather than instability. Overall Fed-LWR balances fast convergence and improved fairness.

VIII. Analytical Discussion

A. Why Layer-wise Re-weighting Works

Layer-wise CKA captures structural differences in learned representations. By increasing influence where global representations diverge from a client's local features, aggregation corrects layer-specific mismatches without altering well-aligned layers, which preserves shared low-level features and focuses correction on semantic differences.

B. Trade-offs

Fed-LWR emphasizes fairness and equitable performance across clients. This comes at computional cost for per-layer similarity computation (CKA), but the benefits in stability and cross-client consistency often justify the overhead in medical deployments where fairness is critical.

IX. Limitations

Fed-LWR has several limitations:

- **Privacy exposure:** Layer-wise statistics may leak information unless combined with differential privacy or secure aggregation.
- **Compute overhead:** Per-layer similarity estimation (CKA) increases computation at clients and server.
- **Task generalization:** The method is optimized for segmentation and may require adaptation for classification or detection tasks.
- **Metric dependency:** Relying on a single similarity metric (CKA) might not capture all types of domain shift; combining multiple metrics is a potential extension.

X. Hier-Fed-LWR: Hierarchical Extension

A. Motivation

Large-scale federations (many hospitals, multiple regions) introduce bandwidth, privacy, and stability concerns. A hierarchical design reduces communication overhead and leverages regional aggregation to stabilize global updates.

B. Architecture and Advantages

- **Multi-level FL:** Hospitals $\rightarrow$ Regional Aggregators $\rightarrow$ Global Server.
- **Enhanced fairness:** Layer-wise re-weighting applied locally and regionally encourages uniform performance across hospitals.
- **Privacy preservation:** Raw activations remain local; only low-dimensional summaries or divergence metrics are communicated.
- **Communication efficiency:** Regional aggregation reduces load on the global server.
- **Scalability:** Supports larger federations by dividing aggregation responsibilities.

C. Implementation notes

Hier-Fed-LWR performs Fed-LWR at the hospital-regional level, then aggregates regionally using analogous layer-wise weighting. Regional summaries reduce global communication while preserving fairness benefits.

XI. Hier-Fed-LWR: Hierarchical Extension

A. Motivation and Architecture

Hier-Fed-LWR extends Fed-LWR to a multi-level federated setup:

- **Multi-Level FL Architecture:** Hospitals $\rightarrow$ Regional Aggregators $\rightarrow$ Global Server.
- **Enhanced Fairness:** Layer-wise re-weighting is applied locally and regionally, improving uniformity across hospitals.
- **Privacy Preservation:** Raw activations never leave hospitals; only summary statistics or divergence metrics are communicated.
- **Communication Efficiency:** Regional aggregation reduces load on the global server.
- **Scalability:** Hierarchical aggregation supports large-scale federations effectively.

XII. Retinal Fundus Imaging

A. Definition

Retinal fundus imaging is a non-invasive method used to capture the interior surface of the eye, including the retina, optic disc, macula, and vasculature.

B. Clinical Importance

- Early detection of diabetic retinopathy, glaucoma, AMD, and hypertensive retinopathy.
- Critical for mass screening and longitudinal monitoring of ocular health.

C. Characteristics

- High inter-patient variability due to illumination, pigmentation, and device type.
- Strong domain shifts across hospitals, cameras, and acquisition protocols.
- Sensitive to noise, blur, and imaging artifacts.

D. Relevance to Federated Learning

- Data siloed across hospitals due to privacy constraints.
- Severe heterogeneity makes centralized learning infeasible.
- Ideal for evaluating robustness of FL algorithms under domain shift.

XIII. Behavior of Standard Fed-LWR Across Domains

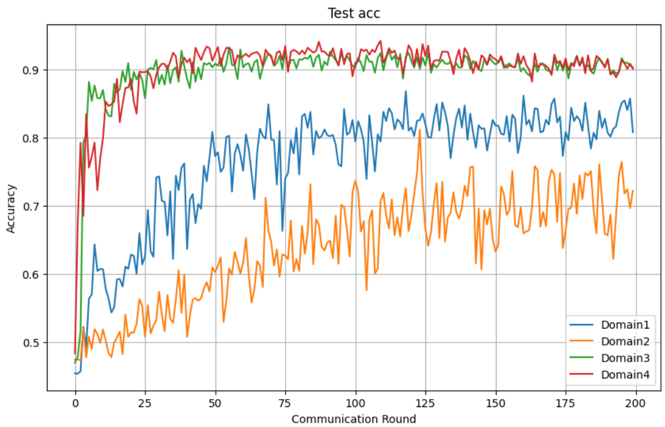


Fig. 4: Test accuracy for various Domains trained on Fed-LWR

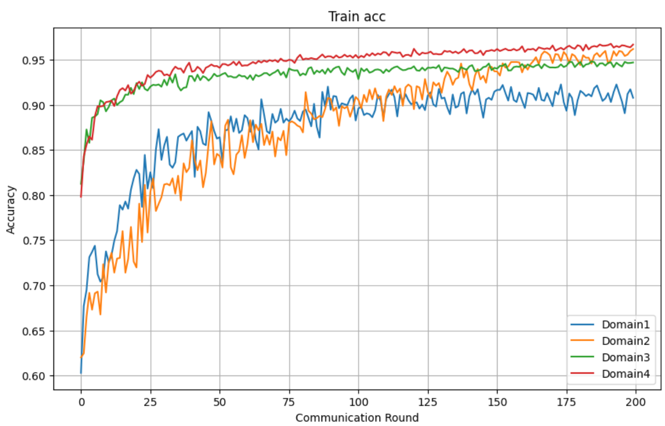


Fig. 5: Train accuracy for various Domains trained on Fed-LWR

Fig. 6: Test Loss for various Domains trained on Fed-LWR

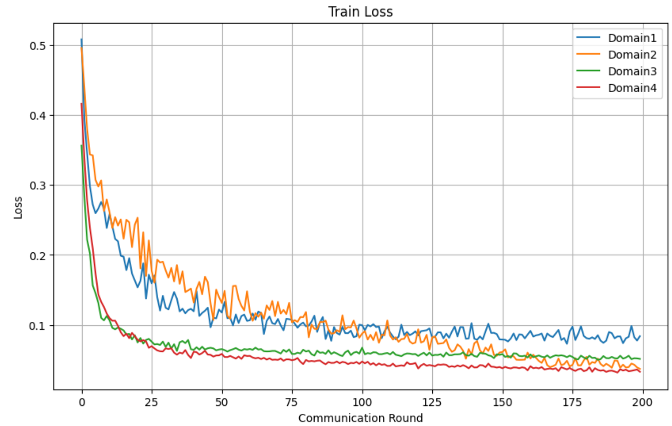


Fig. 7: Train Loss for various Domains trained on Fed-LWR

A. Training Behavior

- Rapid loss reduction during the first 20–30 rounds across all domains.
- Domains 3 and 4 converge fastest with stable curves.
- Domains 1 and 2 converge slower and exhibit higher variance.
- Domain 4 reaches ∼96% training accuracy; Domain 3 achieves 94–95%.
- Domains 1 and 2 eventually exceed 90%, but with significant noise.

B. Generalization Behavior

- Domains 3 and 4 achieve strongest test accuracy (90–92%).
- Domain 1 stabilizes around 82–85%.
- Domain 2 suffers fluctuations and high test loss.

Standard Fed-LWR struggles in highly shifted domains.

XIV. Fed-LWR vs. Hier-Fed-LWR Across Domains

XV. Fed-LWR vs. Hier-Fed-LWR Across Domains

A. Well-Aligned Domains (Clients 3 & 4)

Training:

- Both methods improve rapidly to 92–95%.
- Fed-LWR converges faster initially.
- Hier-Fed-LWR provides smoother accuracy trajectories.

Generalization:

- Fed-LWR achieves lower test loss and faster convergence.
- Hier-Fed-LWR is more stable with cleaner test curves.

B. Moderately Shifted Domain (Client 2)

Training:

- Both methods show fluctuations.
- Fed-LWR reaches higher peaks but is unstable.
- Hier-Fed-LWR provides smoother and steady performance.

Generalization:

- Final test losses nearly identical.
- Fed-LWR oscillates; Hier-Fed-LWR remains stable.

(a) Client 3 Test Accuracy (b) Client 4 Test Accuracy (c) Client 3 Test Loss (d) Client 4 Test Loss

(e) Client 2 Test Loss (f) Client 2 Test Accuracy (g) Client 1 Test Loss (h) Client 1 Test Accuracy

Fig. 8: Fed-LWR vs Hier-Fed-LWR performance across multiple domain conditions.

C. Highly Shifted Domain (Client 1)

Training:

- Fed-LWR achieves higher peaks but with strong oscillations.
- Hier-Fed-LWR improves gradually with reduced instability.

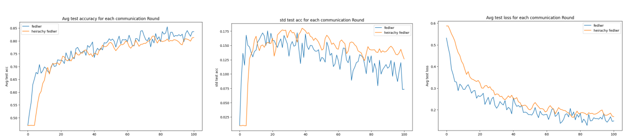
Generalization:

- Fed-LWR lowers test loss quickly but fluctuates heavily.
- Hier-Fed-LWR offers smoother convergence and more reliable behavior.

Summary:

- Fed-LWR excels on well-aligned domains (speed, low loss).
- Hier-Fed-LWR excels on heterogeneous domains (stability).

XVI. Accuracy, Stability, and Loss Comparison



(a) Avg Test Acc (b) Std Test Acc (c) Avg Test Loss

Fig. 9: Accuracy and Stability comparison between Fed-LWR and Hier-Fed-LWR.

A. Accuracy & Stability

- **Lower Variation:** Fed-LWR shows larger cross-client variation; Hier-Fed-LWR stabilizes earlier.
- **Smoothness:** Hier-Fed-LWR yields cleaner accuracy curves.
- **Faster Rise:** Fed-LWR increases accuracy faster in early rounds.
- **Final Accuracy:** Fed-LWR slightly surpasses Hier-Fed-LWR.
- **Consistency:** Hier-Fed-LWR provides more stable accuracy across heterogeneous clients.

B. Test Loss & Convergence

- Fed-LWR converges faster and reduces loss earlier.
- Fed-LWR maintains lowest final loss across domains.
- Hier-Fed-LWR descends smoothly with reduced fluctuations.
- Hier-Fed-LWR more robust under strong domain shift.

Trade-off: Fed-LWR offers speed and peak accuracy, whereas Hier-Fed-LWR provides stability and robustness.

XVII. Final Summary and Best Dice Scores

- **Accuracy:** Fed-LWR achieves slightly higher final accuracy in aligned domains.
- **Stability:** Hier-Fed-LWR offers smoother accuracy and loss curves.
- **Generalization:** Fed-LWR achieves lower test loss; Hier-Fed-LWR reduces variance.
- **Robustness:** Hier-Fed-LWR performs better under domain shift.
- **Compute Cost:** Fed-LWR is lighter; Hier-Fed-LWR requires more computation.

Best Scores:

- Fed-LWR best avg Dice: **0.861486** at round 195.
- Hier-Fed-LWR best avg Dice: **0.818923** at round 75.

XVIII. Running Fed-LWR and Hier-Fed-LWR Notebooks

The official implementation of Fed-LWR and Hier-Fed-LWR is provided in the form of Kaggle notebooks. This section describes how to execute the notebooks end-to-end using the RIF Fundus dataset, including preprocessing the images, updating the configuration paths, and running the training pipeline.

A. Preparing the Dataset

The RIF Fundus dataset must first be preprocessed into the federated `.pyc` format. The repository provides a preprocessing utility (`preprocess_rif.ipynb`) that performs image normalization, client-wise partitioning, and serialization. The notebook instructs the user to upload the raw fundus folders to Kaggle and run the preprocessing cell, which produces a directory containing the client-specific `.pyc` files. These files are then used as the input dataset for both Fed-LWR and Hier-Fed-LWR training.

B. Updating Paths in the Notebook

Each notebook contains a configuration cell with a `get_args()` function. This function defines the paths used for reading the dataset and saving the model outputs. Users must update the default paths so that the notebook reads the preprocessed `.pyc` files generated in the previous step. The only modifications required are:

- Set the dataset path to the folder containing the `.pyc` files in `/kaggle/working/`.
- Set the output directory to a writable location such as `/kaggle/working/results/`.

After these changes, the remaining notebook cells automatically load the arguments and propagate the paths throughout the training pipeline.

C. Executing the Fed-LWR Notebook

Once the dataset paths are updated, the Fed-LWR notebook can be executed from start to end. The notebook performs local client updates, computes layer-wise CKA similarity between client representations and the global anchor model, and applies the Fed-LWR aggregation rule. Progress plots, checkpoints, and evaluation logs are saved automatically in the specified output directory.

D. Executing the Hier-Fed-LWR Notebook

The Hier-Fed-LWR notebook extends the same workflow by introducing regional aggregation. After updating the dataset and output paths in the configuration cell, the notebook can be executed cell-by-cell. The notebook handles hospital-level updates, regional server aggregation, and global synchronization. It also reports metrics for stability, accuracy variance, and robustness under domain shift.

E. Kaggle Environment Notes

Kaggle provides GPU-enabled sessions that are sufficient for running both notebooks. No additional setup is required beyond uploading the dataset and the preprocessing file. The notebooks automatically create client partitions, initialize models, run the training loops, and generate plots for accuracy, loss, and cross-client variance.

This notebook-based workflow allows full reproducibility without requiring command line execution or local environment configuration.

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